LANDFORMS BY RUNNING WATER / INDIA DRAINAGE AND INTERLINKING

BASIN ANATOMY → RIVER-WORK ENGINE → LONG-PROFILE CONTROLS → MEANDER-FLOODPLAIN LOOP → DRAINAGE-PATTERN MATRIX → DRAINAGE-EVOLUTION FIREWALL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 BASIN ANATOMY

BASIN ANATOMY | DRAINAGE DIVIDE | RUNOFF ENTERS TRIBUTARIES | TRUNK STREAM | BASIN RULE | UPSTREAM CHANGE TRAVELS DOWNSTREAM

DEFINITION / COMPONENTS
- RIDGELINE → watershed / drainage divide
- SLOPES → runoff enters tributaries

TRIGGER / MECHANISM
- TRIBUTARIES → trunk stream → outlet
- BASIN RULE → upstream change travels downstream

EFFECT / INTERVENTION
- RIDGELINE → watershed / drainage divide
- SLOPES → runoff enters tributaries

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRIBUTARIES → trunk stream → outlet; BASIN RULE → upstream change travels downstream.

01

STAGE 01 RIVER-WORK ENGINE

RIVER-WORK ENGINE | ENERGY + DISCHARGE | EROSION CAPACITY | LOAD SIZE + VOLUME | TRANSPORT DEMAND | COMPETENCE FALLS | DEPOSITION BEGINS | BED FORM ALTERS VELOCITY AND LOAD

ENERGY + DISCHARGE → erosion capacity → LOAD SIZE + VOLUME → transport demand → COMPETENCE FALLS → deposition begins → FEEDBACK → bed form alters velocity and load

DEFINITION / COMPONENTS
- ENERGY + DISCHARGE → erosion capacity
- LOAD SIZE + VOLUME → transport demand

TRIGGER / MECHANISM
- COMPETENCE FALLS → deposition begins
- FEEDBACK → bed form alters velocity and load

EFFECT / INTERVENTION
- ENERGY + DISCHARGE → erosion capacity
- LOAD SIZE + VOLUME → transport demand

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FEEDBACK → bed form alters velocity and load.

02

STAGE 02 LONG-PROFILE CONTROLS

LONG-PROFILE CONTROLS | BASE LEVEL | POTENTIAL-ENERGY CHANGE | ROCK + DISCHARGE | LOCAL INCISION | ABRASION, TRANSPORT AND STORAGE | GRADED CURVE | MOVING DYNAMIC BALANCE

UPLIFT / BASE LEVEL → potential-energy change → ROCK + DISCHARGE → local incision → SEDIMENT → abrasion, transport and storage → GRADED CURVE → moving dynamic balance

DEFINITION / COMPONENTS
- UPLIFT / BASE LEVEL → potential-energy change
- ROCK + DISCHARGE → local incision

TRIGGER / MECHANISM
- SEDIMENT → abrasion, transport and storage
- GRADED CURVE → moving dynamic balance

EFFECT / INTERVENTION
- UPLIFT / BASE LEVEL → potential-energy change
- ROCK + DISCHARGE → local incision

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SEDIMENT → abrasion, transport and storage; GRADED CURVE → moving dynamic balance.

03

STAGE 03 MEANDER-FLOODPLAIN LOOP

MEANDER-FLOODPLAIN LOOP | OUTER BANK | FASTER FLOW | INNER BANK | SLOWER FLOW | POINT BAR | NECK CUTOFF | OX-BOW LAKE

OUTER BANK → faster flow → erosion → INNER BANK → slower flow → point bar → NECK CUTOFF → ox-bow lake → OVERBANK SILT → floodplain + natural levee

DEFINITION / COMPONENTS
- OUTER BANK → faster flow → erosion
- INNER BANK → slower flow → point bar

TRIGGER / MECHANISM
- NECK CUTOFF → ox-bow lake
- OVERBANK SILT → floodplain + natural levee

EFFECT / INTERVENTION
- OUTER BANK → faster flow → erosion
- INNER BANK → slower flow → point bar

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INNER BANK → slower flow → point bar.

03

STAGE 03

MEANDER-FLOODPLAIN LOOP

MEANDER-FLOODPLAIN LOOP · OUTER BANK · FASTER FLOW · INNER BANK · SLOWER FLOW · POINT BAR · NECK CUTOFF · OX-BOW LAKE

OUTER BANK → faster flow → erosion → INNER BANK → slower flow → point bar → NECK CUTOFF → ox-bow lake → OVERBANK SILT → floodplain + natural levee

DEFINITION / COMPONENTS

- OUTER BANK → faster flow → erosion
- INNER BANK → slower flow → point bar

TRIGGER / MECHANISM

- NECK CUTOFF → ox-bow lake
- OVERBANK SILT → floodplain + natural levee

EFFECT / INTERVENTION

- OUTER BANK → faster flow → erosion
- INNER BANK → slower flow → point bar

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INNER BANK → slower flow → point bar.

04

STAGE 04

DRAINAGE-PATTERN MATRIX

DRAINAGE-PATTERN MATRIX · UNIFORM RESISTANCE · ALTERNATING RESISTANT BANDS · JOINTS AND FAULTS · OUTWARD HIGH · INWARD BASIN

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
DENDRITIC → uniform resistance	RECTANGULAR → joints and faults	DENDRITIC → uniform resistance	RECTANGULAR → joints and faults
TRELLIS → alternating resistant bands	RADIAL / CENTRIPETAL → outward high / inward basin	TRELLIS → alternating resistant bands	RADIAL / CENTRIPETAL → outward high / inward basin

CLASSIFICATION

- DENDRITIC → uniform resistance
- TRELLIS → alternating resistant bands

DIAGNOSTIC BASIS

- RECTANGULAR → joints and faults
- RADIAL / CENTRIPETAL → outward high / inward basin

PROCESS LINK

- DENDRITIC → uniform resistance
- TRELLIS → alternating resistant bands

EXAM TRAP

- RECTANGULAR → joints and faults
- RADIAL / CENTRIPETAL → outward high / inward basin

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RADIAL / CENTRIPETAL → outward high / inward basin.

05

STAGE 05

DRAINAGE-EVOLUTION FIREWALL

DRAINAGE-EVOLUTION FIREWALL · RIVER OLDER THAN UPLIFT · COVER PATTERN INHERITED BELOW · HEADWARD EROSION DIVERTS NEIGHBOUR · FIELD CUES · ELBOW-WIND GAP

ANTECEDENT → river older than uplift → SUPERIMPOSED → cover pattern inherited below → CAPTURE → headward erosion diverts neighbour → FIELD CUES → gorge / discordance / elbow-wind gap

DEFINITION / COMPONENTS

- ANTECEDENT → river older than uplift
- SUPERIMPOSED → cover pattern inherited below

TRIGGER / MECHANISM

- CAPTURE → headward erosion diverts neighbour
- FIELD CUES → gorge / discordance / elbow-wind gap

EFFECT / INTERVENTION

- ANTECEDENT → river older than uplift
- SUPERIMPOSED → cover pattern inherited below

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FIELD CUES → gorge / discordance / elbow-wind gap.

06

STAGE 06

INDIA RIVER-SYSTEM CONTRAST

INDIA RIVER-SYSTEM CONTRAST · HIGH RELIEF, MIXED PERENNIAL INPUTS · OLD SHIELD, MONSOON-DOMINANT REGIME · LONG RIVERS AND MAJOR DELTAS · NARMADA-TAPI TROUGHS + SHORT GHATS STREAMS

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
HIMALAYAN → high relief, mixed perennial inputs	EAST → long rivers and major deltas	HIMALAYAN → high relief, mixed perennial inputs	EAST → long rivers and major deltas
PENINSULAR → old shield, monsoon-dominant regime	WEST → Narmada-Tapi troughs + short Ghats streams	PENINSULAR → old shield, monsoon-dominant regime	WEST → Narmada-Tapi troughs + short Ghats streams

SYSTEM / DEFINITION

- HIMALAYAN → high relief, mixed perennial inputs
- PENINSULAR → old shield, monsoon-dominant regime

PROCESS / MECHANISM

- EAST → long rivers and major deltas
- WEST → Narmada-Tapi troughs + short Ghats streams

SPATIAL / INDIA PATTERN

- HIMALAYAN → high relief, mixed perennial inputs
- PENINSULAR → old shield, monsoon-dominant regime

HAZARD / LIMIT / RESPONSE

- EAST → long rivers and major deltas
- WEST → Narmada-Tapi troughs + short Ghats streams

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WEST → Narmada-Tapi troughs + short Ghats streams.

07

STAGE 07

SOURCE-COURSE-MOUTH RAIL

SOURCE-COURSE-MOUTH RAIL · TIBETAN HEADWATERS · ARABIAN SEA · DEVPRAYAG NAME · BENGAL DELTA · JOINED DELTA

06

STAGE 06 INDIA RIVER-SYSTEM CONTRAST

INDIA RIVER-SYSTEM CONTRAST · HIGH RELIEF, MIXED PERENNIAL INPUTS · OLD SHIELD, MONSOON-DOMINANT REGIME · LONG RIVERS AND MAJOR DELTAS · NARMADA-TAPI TROUGHS + SHORT GHATS STREAMS

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
HIMALAYAN → high relief, mixed perennial inputs	EAST → long rivers and major deltas	HIMALAYAN → high relief, mixed perennial inputs	EAST → long rivers and major deltas
PENINSULAR → old shield, monsoon-dominant regime	WEST → Narmada-Tapi troughs + short Ghats streams	PENINSULAR → old shield, monsoon-dominant regime	WEST → Narmada-Tapi troughs + short Ghats streams

SYSTEM / DEFINITION
- HIMALAYAN → high relief, mixed perennial inputs
- PENINSULAR → old shield, monsoon-dominant regime

PROCESS / MECHANISM
- EAST → long rivers and major deltas
- WEST → Narmada-Tapi troughs + short Ghats streams

SPATIAL / INDIA PATTERN
- HIMALAYAN → high relief, mixed perennial inputs
- PENINSULAR → old shield, monsoon-dominant regime

HAZARD / LIMIT / RESPONSE
- EAST → long rivers and major deltas
- WEST → Narmada-Tapi troughs + short Ghats streams

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WEST → Narmada-Tapi troughs + short Ghats streams.

07

STAGE 07 SOURCE-COURSE-MOUTH RAIL

SOURCE-COURSE-MOUTH RAIL · TIBETAN HEADWATERS · ARABIAN SEA · DEVPRAYAG NAME · BENGAL DELTA · JOINED DELTA

INDUS → Tibetan headwaters → Arabian Sea

GANGA → Devprayag name → Bengal delta → BRAHMAPUTRA → Tsangpo-Siang → joined delta → PENINSULA → Godavari-Krishna-Kaveri / Narmada-Tapi

SYSTEM / DEFINITION
- INDUS → Tibetan headwaters → Arabian Sea
- GANGA → Devprayag name → Bengal delta

PROCESS / MECHANISM
- BRAHMAPUTRA → Tsangpo-Siang → joined delta
- PENINSULA → Godavari-Krishna-Kaveri / Narmada-Tapi

SPATIAL / INDIA PATTERN
- INDUS → Tibetan headwaters → Arabian Sea
- GANGA → Devprayag name → Bengal delta

HAZARD / LIMIT / RESPONSE
- BRAHMAPUTRA → Tsangpo-Siang → joined delta
- PENINSULA → Godavari-Krishna-Kaveri / Narmada-Tapi

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BRAHMAPUTRA → Tsangpo-Siang → joined delta; PENINSULA → Godavari-Krishna-Kaveri / Narmada-Tapi.

08

STAGE 08 DELTA BUDGET

DELTA BUDGET · BASIN SEDIMENT DELIVERY · WAVES, TIDES AND CYCLONE SCOUR · COMPACTION + RELATIVE SEA LEVEL · DISTRIBUTARY MOBILITY OR CONFINEMENT

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
INPUT → basin sediment delivery	VERTICAL → compaction + relative sea level	INPUT → basin sediment delivery
REMOVAL → waves, tides and cyclone scour	CHANNELS → distributary mobility or confinement	REMOVAL → waves, tides and cyclone scour

DEFINITION / COMPONENTS
- INPUT → basin sediment delivery
- REMOVAL → waves, tides and cyclone scour

TRIGGER / MECHANISM
- VERTICAL → compaction + relative sea level
- CHANNELS → distributary mobility or confinement

EFFECT / INTERVENTION
- INPUT → basin sediment delivery
- REMOVAL → waves, tides and cyclone scour

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERTICAL → compaction + relative sea level; CHANNELS → distributary mobility or confinement.

09

STAGE 09 TRANSFER VIABILITY TEST

TRANSFER VIABILITY TEST · SEASONAL WATER AFTER EXISTING CLAIMS · DEMAND, SOIL, AQUIFER AND DELIVERY FIT · LOSS, ENERGY, ECOLOGY AND PEOPLE · RECHARGE, EFFICIENCY, REUSE, OPERATIONS

DONOR → seasonal water after existing claims

RECIPIENT → demand, soil, aquifer and delivery fit → CORRIDOR → loss, energy, ecology and people → ALTERNATIVES → recharge, efficiency, reuse, operations

DEFINITION / COMPONENTS
- DONOR → seasonal water after existing claims
- RECIPIENT → demand, soil, aquifer and delivery fit

TRIGGER / MECHANISM
- CORRIDOR → loss, energy, ecology and people
- ALTERNATIVES → recharge, efficiency, reuse, operations

EFFECT / INTERVENTION
- DONOR → seasonal water after existing claims
- RECIPIENT → demand, soil, aquifer and delivery fit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RECIPIENT → demand, soil, aquifer and delivery fit.

09

STAGE 09 TRANSFER VIABILITY TEST

TRANSFER VIABILITY TEST | SEASONAL WATER AFTER EXISTING CLAIMS | DEMAND, SOIL, AQUIFER AND DELIVERY FIT | LOSS, ENERGY, ECOLOGY AND PEOPLE | RECHARGE, EFFICIENCY, REUSE, OPERATIONS

DONOR → seasonal water after existing claims

RECIPIENT → demand, soil, aquifer and delivery fit → CORRIDOR → loss, energy, ecology and people → ALTERNATIVES → recharge, efficiency, reuse, operations

DEFINITION / COMPONENTS
- DONOR → seasonal water after existing claims
- RECIPIENT → demand, soil, aquifer and delivery fit

TRIGGER / MECHANISM
- CORRIDOR → loss, energy, ecology and people
- ALTERNATIVES → recharge, efficiency, reuse, operations

EFFECT / INTERVENTION
- DONOR → seasonal water after existing claims
- RECIPIENT → demand, soil, aquifer and delivery fit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RECIPIENT → demand, soil, aquifer and delivery fit.

10

STAGE 10 KEN-BETWA BOUNDARY

KEN-BETWA BOUNDARY | KEN BASIN TO BETWA BASIN | OFFICIAL MATERIAL SAYS UNDER IMPLEMENTATION | PANNA CONNECTIVITY + DOWNSTREAM FLOW | STATUS IS NOT MEASURED OUTCOME

DIRECTION → Ken basin to Betwa basin

STATUS → official material says under implementation → LANDSCAPE → Panna connectivity + downstream flow → RULE → status is not measured outcome

DEFINITION / COMPONENTS
- DIRECTION → Ken basin to Betwa basin
- STATUS → official material says under implementation

TRIGGER / MECHANISM
- LANDSCAPE → Panna connectivity + downstream flow
- RULE → status is not measured outcome

EFFECT / INTERVENTION
- DIRECTION → Ken basin to Betwa basin
- STATUS → official material says under implementation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LANDSCAPE → Panna connectivity + downstream flow; RULE → status is not measured outcome.

11

STAGE 11 BASIN ANSWER SPINE

BASIN ANSWER SPINE | DEFINE + MAP | BASIN AND DIRECTION | PROCESS, TIMING AND SEDIMENT | BENEFIT MECHANISM AGAINST LIMITS | SELECTIVE, ADAPTIVE AND BASIN-WIDE

DEFINE + MAP → basin and direction → EXPLAIN → process, timing and sediment → ASSESS → benefit mechanism against limits → VERDICT → selective, adaptive and basin-wide

DEFINITION / COMPONENTS
- DEFINE + MAP → basin and direction
- EXPLAIN → process, timing and sediment

TRIGGER / MECHANISM
- ASSESS → benefit mechanism against limits
- VERDICT → selective, adaptive and basin-wide

EFFECT / INTERVENTION
- DEFINE + MAP → basin and direction
- EXPLAIN → process, timing and sediment

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ASSESS → benefit mechanism against limits; VERDICT → selective, adaptive and basin-wide.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | THE INDUS, GANGA AND BRAHMAPUTRA SYSTEMS CROSS INTERNATIONAL BOUNDARIES | ANTECEDENT DRAINAGE | RIVERS PREDATING THE HIMALAYA THAT MAINTAINED THEIR COURSE BY | RIVER CAPTURE (PIRACY) | A MORE VIGOROUS RIVER, THROUGH HEADWARD EROSION, BEHEADS A | INDO-BRAHM HYPOTHESIS (KHULLAR) | A MIGHTY FORMER RIVER ALONG THE HIMALAYAN FOREDEEP WAS

OPTIONAL PROCESS DEPTH
- The Indus, Ganga and Brahmaputra systems cross international boundaries and are mostly perennial.
- Antecedent drainage: rivers predating the Himalaya that maintained their course by down-cutting as the range uplifted (deep transverse gorges).
- River capture (piracy): a more vigorous river, through headward erosion, beheads a neighbouring river — a key process in Himalayan drainage evolution.
- Indo-Brahm hypothesis (Khullar): a mighty former river along the Himalayan foredeep was

DATA / INSTITUTION LIMIT
- dismembered by successive captures into today's Indus, Ganga and Brahmaputra systems.
- Most large Peninsular basins drain east; Narmada and Tapi flow west through structurally controlled
- Objective: transfer water within Bundelkhand for irrigation, drinking water and power; quote
- Concerns: submergence and fragmentation around Panna Tiger Reserve, downstream-flow effects,

CONTEMPORARY PARALLEL
- Institutions: Central Water Commission, Ken-Betwa Link Project Authority; other links for comparison — Par-Tapi-Narmada, Damanganga-Pinjal.
- Himalayan systems are mostly perennial; Peninsular regimes are more monsoon-seasonal and occupy an older landscape.
- Antecedent river = older than the mountain it cuts across (deep gorge).
- River capture driven by headward erosion (Khullar: Indo-Brahm captures).

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels